Ramu Vemanna's Lab



Home

In Ramu's lab, we explore plants' stress-adaptation mechanisms to develop climate-resilient, high-nutritional-value plants.

Research

Translation mechanisms that contribute to plant development and drought-stress adaptation. Drought stress tolerance is multigenic. Protein synthesis is affected under drought stress, which, in turn, influences cellular metabolic activities required for optimum functioning. Ribosomes are responsible for protein synthesis. However, ribosome biogenesis is a highly orchestrated process involving many proteins that support cellular translation. We want to unravel how translational mechanisms regulate the adaptation abilities of plants under changing environmental conditions.

Areas of interest

Protein translational mechanisms in plants.

Extra-ribosomal functions of ribosomal proteins.

Ribosome biogenesis factors in plant growth and development.

Regulatory proteins of ribosome biogenesis.

Drought stress adaptation of plants.

Our research focuses on understanding the basis of plant adaptation to drought to improve the productivity of our crops. Under stress conditions, the acquired tolerant traits play a crucial role as a measure of cellular level tolerance. Emphasis is on Ribosomal RNA (rRNA) processing mechanisms, ribosome biogenesis and their assembly factors. Functional characterisation of the genes involved in these mechanisms and exploring their extra-ribosomal functions and relevance in growth,

development and abiotic stress tolerance. Further, we want to unravel how translational regulation contributes to acclimation processes in changing climatic conditions. Identifying the relevant cis-elements on ribosomal RNA genes and ribosomal proteins encoding gene promoters and transcription factors involved in their expression and regulation of translational activity has phenomenal significance.

We are also working on developing strategies to improve resistance against bacterial leaf blight disease in rice-causing pathogen Xanthomonas oryzae pv oryzae. We use functional and chemical genomic approaches to improve plant health.

People



Shobna Yadav, PhD Scholar

Shobhna is trying to unravel rRNA processing, a step in ribosome biogenesis required for plant growth and development. Other than that, you will find her untangling any other interesting problem.



Libin Thomas, PhD Scholar

Libin is studying the changes in Ribosome Composition and Functional Insight in Rice under drought conditions, and is also interested in deciphering the functional significance of the RPL10-interacting ABC protein.



Manisha Yadav, PhD Scholar

Manisha is trying to understand the molecular mechanisms of ribosome heterogeneity and the translational regulation underlying plant adaptation to abiotic stresses, with a particular emphasis on drought tolerance in rice. She is particularly interested in understanding how these regulatory mechanisms coordinate plant growth, development, and adaptation under changing environmental conditions.



Sathya Seelan, PhD Scholar

Sathya works on Dry-seeded rice – Engineering organellar translational associated factors to improve seed germination, seedling vigor, and drought tolerance. In his leisure time, he enjoys cooking.



Shubham Singh, PhD Scholar

Shubham's research focuses on elucidating the molecular mechanisms underlying rice responses to biotic and abiotic stresses using functional genomics, genome editing, and multi-omics approaches to enhance crop resilience. Apart from research work, Shubham likes singing.



Suchismita Satapathy, PhD Scholar

Suchi is on a quest to unravel the “how” and “why” of rRNA in rice development, growth, and survival, with a focus on plant adaptation and resilience. Beyond the lab, she continues to explore the “how” and “why” of the universe, searching for answers to endless questions that spark her curiosity.



Dana Basheer, PhD scholar

Dana is currently investigating the molecular mechanism underlying seed dormancy in rice under drought stress, focusing on the roles of ABA signaling and translational regulation. Apart from her research work, she enjoys watching movies and baking.



Avani Vasudeva, PhD Scholar

Avani is currently working to elucidate the interactions and functional roles of polyubiquitin proteins in rice, with a focus on their potential involvement in stress regulation and cellular processes. Beyond Research, Avani loves to explore nature.



Bariz Ali, PhD Scholar

Bariz works on solving viral infections in potatoes, with a particular interest in understanding virus–plant interactions. He sought to understand the molecular aspects of viral infection and potato plants' responses to viral stress. Outside the laboratory, He enjoys playing cricket and badminton.



Swetannita Chattopadyay, Senior Research Fellow

Swetannita did her PhD in"miR160 and its targets ARF role in root system development under nitrogen deficiency in Brassica juncea." Now she is working on ribosomal RNA (rRNA) diversity, the functional relevance of rRNA processing. Beyond the bench, she is a trained Odissi classical dancer. She loves painting and creating mandala artwork with a keen interest in abstract art.



Umang Kumar Dhandanya, Project Associate

Umang is working on the genetic dissection of rancidity and goitrogenic c-glycosyl flavone biosynthesis genes to enhance the shelf-life and nutritional quality of pearl millet flour. He loves singing, acting, and playing chess.



Orishma Parida, Project Associate

Orishma is currently working on “An integrated cell wall biochemistry-based strategy for promoting the resilience and circularity of rice cultivation.” She is trying to understand the molecular and regulatory mechanisms that contribute to plant resilience, particularly in rice. Outside the lab, she enjoys practising Odissi classical dance and Hindustani classical music.



Dr Babitha K C, EIR Post-doctoral Fellow,

Dr. Babitha works to develop next-generation agrochemicals that improve plant physiological processes by designing small molecules that target relevant proteins to enhance carbon capture, photosynthesis, and crop productivity. The objective is to translate lab discoveries to field applications.

Patents / Technologies

1. **A Pro-Transpirants Class Combination** Ramu S Vemanna, PA, Vanitha, NV, Preethi (23 August 2021) Indian Patent Application No. 202111038068

The invention relates to the field of agrochemical compositions that improve plant productivity. The objective of the invention is to develop a combination of agrochemicals that enhances transpiration and increases plant yield under drought conditions. Under drought conditions, crop productivity declines due to the accumulation of Abscisic acid (ABA). The present invention discloses a pro-transpirants class combination for downregulating the transcription factor, which reduces the accumulation of ABA in rice, enhances transpiration, and increases the yield of the plant

2. **Transgene-free, inplanta-mediated delivery of RNPS to edit genes in plants** Shankar Mavinamar, Dr. Sneh Lata Singla Pareek, Dr. Avinash Bajaj, Dr. Ramu S. Vemanna Indian Patent Application No. 202511006944 dated 28 January 2025
3. **Novel Small Molecule Modulators of CRISPR-Cas9 Activity and Method Thereof** A. Rode, K. Shinde, D. Ojha, R. Vemanna Patent Number:539344 (27 May 2024)
4. **Pro-transpirants to increase crop productivity**. Babitha KC, Veena HM, and Ramu S Vemanna (7 February 2022) Patent Granted: no. 550300 (13/09/2024)

The present invention's emphasis is to develop a combination that increases transpiration and yield in plants, even under stressful conditions. We developed a formulation/combination that keeps the stomata open and enhances transpiration. The formulations inhibit transporter function and were found to be effective in rice, wheat, cowpea, and mungbean (Greengram) for enhancing yield.

5. **RNP complex for crop protection against bacterial diseases** Pal G, Bajaj A, Bhattacharjee S, Vemanna RS (7 March 2023) 8 Indian patent application number – 202311015285, Patent No. 560014, granted date 12.02.2025

The present invention provides a nanopolymer-RNP complex to deliver RNP complex. Particularly, the invention provides a nanopolymer-RNP complex targeting Xanthomonas oryzae pv oryzae (Xoo) bacterial hrpX and hrpG genes that protects from bacterial blight disease on rice and Rolstonia solanacearum (RS) hrpX and hrpG in potato that protects from bacterial wilt, and Psuedomonas syringae pv tomato (DC3000) (PSt) HopP1 protecting from bacterial speck on Arabidopsis.

6. **Synthetic and natural molecules for crop protection against plant diseases caused by bacteria, and nematode and virus**. Garima Pal, Ramu S Vemanna, 202411062781, TEMP/E-1/72958/2024-DEL, (Provisional patent filed – filed for complete filing).

The invention, in general, relates to the field of plant protection against broad spectrum of diseases. More particularly, the present invention relates to Synthetic and natural molecules for crop protection against plant diseases caused by bacteria, nematode and virus and method thereof.

PUBLICATION:-

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2. Yadav S, Mathew K, Singh S, Biswas A, Deshpande S, Kumari C, Reddy SK, Wang K, Maiti TK, Mysore KS, Vemanna RS. Ribosome Processing Factor-2 Interacts with RPL10A to Regulate Selective Translation during Plant Immunity and Drought Stress. bioRxiv. 2026 Mar 13:2026-03.
3. Yadav S, Preethi V, Dadi S, Seth CS, G K, Chandrashekar BK, Vemanna RS. Small chemical molecules regulating the phytohormone signalling alter the plant's physiological processes to improve stress adaptation, growth and productivity.Physiol Mol Biol Plants. 2024 Oct;30(10):1593-1610. doi: 10.1007/s12298-024-01514-w. Epub 2024 Oct 3. PMID: 39506995
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PROJECTS:-

1. Studying the ribosomal RNA (rRNA)diversity, functional relevance of rRNA processing factor 2 (RPF2) in ribosome biogenesis, root, shoot development and drought stress tolerance in rice
2. Genetic dissection of rancidity and goitrogenic c-glycosyl flavone biosynthesis genes to enhance the shelf-life and nutritional quality of pearl millet flour
3. An integrated cell wall biochemistry-based strategy for promoting the resilience and circularity of rice cultivation (Indo-Sweden, Bilateral)
4. Nanogel-mediated Gene Editing (CRISPR/Cas9) Technologies to Improve Crop Protection against Bacterial Leaf Blight in Rice

5. Influence of drought stress on ribosomes and protein synthesis and understanding the functional relevance of Ribosomal Protein L10 in rice.
6. Identification and functional characterization of genes involved in protein translation, degradation, and develop inhibitors to understand bacterial pathogenicity in rice
7. Disruption of genome integrity to create genetic variability by editing (Using CRISPR Cas9) the genes associated with DNA mismatch repair and characterization of their relevance in crop improvement
8. BRIC-BIRAC Post Doc Entrepreneur-in-Residence (EIR) program An agrochemical inhibiting ABA biosynthesis proteins or anion channels that regulate stomatal opening inspite of ABA accumulation

Join US-

Interested in Postdoctoral fellowships, DBT-RA, NPDF, DBT-BioCARE, WOSA, etc. scheme, please write me with a one-page statement of Interest that aligns with my lab focus and send me two contact details of References.

Alumni

- Dr. Garima Pal, PhD Scholar 2019-2024. Presently a Postdoctoral fellow at the University of Berkeley, USA
- Dr. Akshata Dawane, Postdoctoral Research Fellow · University of Missouri-Kansas City,
- Kesia Mathew, Project -JRF 2024-2026.
- Dr. Punam Sharma Project-SRF, Presently Postdoctoral fellow in NIPGR, New Delhi.
- Devika project asspciate
- Supriya Singh, Project JRF 2024-2026, Presently PhD Student at University of Gottingen, Germany.
- Ms. Veena Mavatkar, Project associate, 2021-2023, Presently PhD student at the Polish Academy of Sciences, Poland.

- Mr. Shubham Agarwal, Project fellow, 2021-2023, Presently PhD student at NIPGR, New Delhi, India.
- Dr Chanchal Kumari, DBT-RA. 2019-2021, Presently, Research associate at IARI, New Delhi, India.
- Dr. Kishore Ingole, Project associate, 2019-202. Presently, a Postdoctoral Fellow at Oxford University, UK.
- Dr. Chethan Chauhan, Research Fellow, 2019- 2020, Presently Research Associate at Regional Center for Biotechnology, Faridabad.
- Ms Shristi Sharma, Project JRF 2021,
- Mr Mukund Mendirata, Project fellow, 2020-2022,
- Ms. Kalai Magal, Project Fellow, 2020-2021.
- Ms. Pooja Yadav, Project Fellow, 2019-2020.

- **Short-term Trainees**.
- Astha Gil
- Mayank Agarwal,
- Sai Chandeeramaulee
- Rashmi Mishra
- Kaiser Parry.
- Mohammad Ahmar Rashid Khan
- Ocean Khajuria
- Labeena Usmani
- Madhu Perla
- Mahi Sharma
- Shreetoma Roy

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